

2nd_Ch5_Example 5-1

Cost information from **eight previously completed parking garage projects** is given in the following table.

以下表格提供了八个已完成的停车场项目的成本信息。

Project	Total cost	No. cars	Unit cost
1	\$1,387,500	150	\$ 9,250
2	896,000	80	11,200
3	1,797,000	120	14,975 ← highest value
4	1,107,000	90	14,975
5	590,400	60	12,300
6	1,903,000	220	8,650 ← lowest value
7	889,000	70	12,700
8	1,615,500	180	8,975
			Total = \$79,034
			Average cost per car = \$ 9,879 ← average value

From Eq. 5.1 the forecast unit cost can be calculated as

$$UC = \frac{\$8,650 + 4(\$9,879) + \$14,975}{6} = \$10,524$$

For a project with 135 cars the estimated cost can be calculated as

$$135 \text{ cars @ } \$10,524 = \$1,420,673$$

E4th_Ch6_Example 6-3

Cost information from **eight previously completed parking garage projects** is given in the following table. **Calculations** are rounded to the nearest **\$10**.

以下表格提供了八个已完成的停车场项目的成本信息。计算结果四舍五入到最接近的**\$10**。

Project	Total cost	No. parking slots	Unit cost	
1	\$1,698,000	145	\$11,710	← lowest value
2	1,100,000	60	18,330	
3	2,547,000	110	23,150	← highest value
4	1,390,000	70	19,860	
5	750,000	50	15,000	
6	3,850,000	200	19,250	
7	896,000	75	11,950	
8	2,025,000	130	15,580	
			Total = \$134,830	
			Average cost per parking slot = \$16,850	← average value

From Eq. 6.1, the forecast unit cost can be calculated as

$$UC = \frac{\$11,710 + 4(\$16,850) + \$23,150}{6}$$

$$UC = \$17,040$$

For a project with 175 parking slots, the estimated cost can be calculated as

$$175 \text{ cars @ } \$17,040 = \$2,982,000$$

2nd_Ch5_Example 5-2

Use the **time and location indices** below to calculate the **forecast cost** for a **building** with **62,700 SF** of **floor area**. The building is to be constructed **3 years from now** in **city B**. A **similar type building** that cost **\$2,945,250** and contained **38,500 SF** was completed **2 years ago** in **city D**.

使用下方的时间和地点指数计算一栋建筑的预计成本，该建筑的地面面积为 **62,700** 平方英尺，计划在 **3** 年后在 **B** 市建造。一个类似的建筑，成本为 **\$2,945,250**，面积为 **38,500** 平方英尺，于 **2** 年前在 **D** 市建成。

Year	Index	Location	Index
3 yr ago	358	City A	1025
2 yr ago	359	City B	1170
1 yr ago	367	City C	1260
Current year	378	City D	1240

An equivalent compound interest can be calculated based on the change in the cost index during the 3-year period:

$$\frac{378}{358} = (1 + i)^3 \rightarrow i = 1.83\%$$

Estimated cost	Previous cost	Time adjustment	Location adjustment	Size adjustment
"	= \$2,945,250	$\times (1 + 0.0183)^3$	$\times (1170/1240)$	$\times (62,700/38,500)$
"	= \$2,945,250	$\times 1.095$	$\times 0.944$	$\times 1.629$
"	= \$4,959,403			

E4th_Ch6_Example 6-4

Use the **time and location indices** below to calculate the **forecast cost** for a **building** with **78,400 square feet** of **floor area**. The building is to be constructed **3 years from now** in **city C**. A **similar type building** that cost **\$3,828,800** and contained **40,400 square feet** was completed **4 years ago** in **city D**.

使用下方的时间和地点指数计算一栋建筑的预计成本，该建筑的地面面积为 **78,400** 平方英尺，计划在 **3** 年后在 **C** 市建造。一个类似的建筑，成本为 **\$3,828,800**，面积为 **40,400** 平方英尺，于 **4** 年前在 **D** 市建成。

Year	Index	Location	Index
4 years ago	4,374	City A	1,646
3 years ago	5,028	City B	1,805
2 years ago	5,199	City C	1,911
1 year ago	5,461	City D	2,091
Current year	5,761	City E	1,876

There is no method to predict future cost escalation with absolute certainty. For this example, a compound interest is calculated based on the trend in cost indices of the past 4 years. Using the compound interest equation $F/P = (1 + i)^n$, an interest rate can be calculated as follows:

$$\frac{5761}{4374} = (1 + i)^4 \rightarrow i = 7.1\%$$

The combined adjustments of costs for time, location, and size can be calculated as follows:

Estimated cost	Previous cost	Time adjustment	Location adjustment	Size adjustment
"	= \$3,828,800	$\times (1.071)^7$	$\times (1911/2091)$	$\times (78,400/40,400)$
"	= \$3,838,800	$\times 1.6163$	$\times 0.9139$	$\times 1.9406$
\$11,004,000				

2nd_Ch5_Example 5-3

This example illustrates the **capital recovery method** for determining **economic feasibility**. Suppose the **feasibility estimate** for a project is **\$7.0M** with an **expected operating life** of **12 years**. **Annual maintenance and operating expenses** are forecast as **\$560K per year**. Using a **10% interest rate**, what **net annual income** must be received to **recover the capital investment** of the project?

此示例展示了资本回收方法来确定项目的经济可行性。假设某项目的可行性估算为**\$7.0M**，预计使用年限为**12**年。每年的维护和运营费用预计为**\$560K**。使用**10%**的利率，项目需要每年获得多少净收入以回收资本投资？

$$\begin{aligned} A &= P \left[\frac{i(1+i)^n}{(1+i)^n - 1} \right] + \$0.56M \\ &= \$7.0M \left[\frac{0.10(1+0.10)^{12}}{(1+0.10)^{12} - 1} \right] + \$0.56M \\ &= \$1.588M \text{ per year} \end{aligned}$$

If the project can be built for \$7.0M, operated for \$560K per year, and earn \$1.5876M per year, the project is economically feasible, neglecting any tax advantages. If the project can be built for less than \$7.0M, or can be built to operate more efficiently than \$560K per year, it is even more economically attractive.

E4th_Ch7_Example 7-1

This example illustrates the **capital recovery method** for determining **economic feasibility**. Suppose the **feasibility estimate** for a project is **\$14.0M** with an **expected operating life** of **15 years**. **Annual maintenance and operating expenses** are forecast as **\$750K per year**. Using a **5% interest rate**, what **net annual income** must be received to **recover the capital investment** of the project?

此示例展示了资本回收方法来确定项目的经济可行性。假设某项目的可行性估算为**\$14.0M**，预计使用年限为**15**年。每年的维护和运营费用预计为**\$750K**。使用**5%**的利率，项目需要每年获得多少净收入以回收资本投资？

$$\begin{aligned} A &= P \left[\frac{i(1+i)^n}{(1+i)^n - 1} \right] + \$0.75M \\ &= \$14.0M \left[\frac{0.05(1+0.05)^{15}}{(1+0.05)^{15} - 1} \right] + \$0.75M \\ &= \$2.10M \text{ per year} \end{aligned}$$

If the project can be built for \$14.0M, operated for \$750K per year, and earn \$2.1M per year, the project is economically feasible, neglecting any tax advantages. If the project can be built for less than \$14.0M, or can be built to operate more efficiently than \$750K per year, it is even more economically attractive.

2nd_Ch5_Example 5-4

This example illustrates the **payback period method** for determining the **economic feasibility** of a project. The **initial investment** for a project is **\$18.0M**. A **net annual profit** of **\$3.5M** is anticipated. Using a **15% desired rate of return** on the investment, what is the **payback period** for the project?

此示例展示了回收期方法来确定项目的经济可行性。该项目的初始投资为**\$18.0M**，预计每年获得净利润为**\$3.5M**。使用 **15%** 的期望投资回报率，该项目的回收期是多少？

$$P = A \left[\frac{(1+i)^n - 1}{i(1+i)^n} \right]$$
$$\$18.0\text{M} = \$3.5\text{M} \left[\frac{(1+0.15)^n - 1}{0.15(1+0.15)^n} \right]$$
$$n \approx 10.5 \text{ years}$$

E4th_Ch7_Example 7-2

This example illustrates the **payback period method** for determining the **economic feasibility** of a project. The **initial investment** for a project is **\$24.0M**. A **net annual profit** of **\$5.0M** is anticipated. Using a **12% desired rate of return** on the investment, what is the **payback period** for the project?

此示例展示了回收期方法来确定项目的经济可行性。该项目的初始投资为**\$24.0M**，预计每年获得净利润为**\$5.0M**。使用 **12%** 的期望投资回报率，该项目的回收期是多少？

$$P = A \left[\frac{(1+i)^n - 1}{i(1+i)^n} \right]$$
$$\$24\text{M} = \$5.0\text{M} \left[\frac{(1+0.12)^n - 1}{0.12(1+.12)^n} \right]$$
$$n = 7.58 \text{ years}$$

2nd_Ch5_Example 5-5

This example illustrates the **rate-of-return method** for determining the **economic feasibility** of a project. An **initial project investment** of **\$1.05M** is being considered for a **5-year study period**. It is anticipated the project will be **sold after the 5-year period** for **\$560K**. Determine the **rate of return** with the anticipated **net profit** shown below.

此示例展示了回报率方法来确定项目的经济可行性。一个初始项目投资为**\$1.05M**，并考虑一个**5年**研究期。预计在**5年后**出售该项目，售价为**\$560K**。计算预期净利润情况下的回报率。

End of year	Net profit (\$1,000)
0	0
1	-350
2	-120
3	+420
4	+735
5	+680

Using Eq. 5.2 to transfer the costs from each year to an equivalent present worth:

Try $i = 15\%$.

$$P = [-\$350(\overset{P/F15-1}{0.8695}) - \$120(\overset{P/F15-2}{0.7562}) + \$420(\overset{P/F15-3}{0.6575}) + \$735(\overset{P/F15-4}{0.5718}) + \$680(\overset{P/F15-5}{0.4972}) + \$560(\overset{P/F15-5}{0.4972})] \times 1,000$$

$\$1.05M > \$0.92M$; therefore try a lower rate of return.

Try $i = 10\%$.

$$P = [-\$350(\overset{P/F10-1}{0.9090}) - \$120(\overset{P/F10-2}{0.8264}) + \$420(\overset{P/F10-3}{0.7513}) + \$735(\overset{P/F10-4}{0.6830}) + \$680(\overset{P/F10-5}{0.6209}) + \$560(\overset{P/F10-5}{0.6209})] \times 1,000$$

$\$1.05M < \$1.17M$; therefore try a higher rate of return.

Try $i = 12\%$.

$$P = [-\$350(\overset{P/F12-1}{0.8929}) - \$120(\overset{P/F12-2}{0.7972}) + \$420(\overset{P/F12-3}{0.7118}) + \$735(\overset{P/F12-4}{0.6355}) + \$680(\overset{P/F12-5}{0.5674}) + \$560(\overset{P/F12-5}{0.5674})] \times 1,000$$

$\$1.05M \approx \$1.06M$; therefore the rate of return is slightly over 12%.

E4th_Ch7_Example 7-3

This example illustrates the **rate-of-return method** for determining the **economic feasibility** of a project. An **initial project investment** of **\$3.0M** is being considered for a **4-year analysis period**. It is anticipated the project will be **sold after the 4-year period** for **\$1.6M**. Determine the **rate of return** with the anticipated **net profit** shown below.

此示例展示了**回报率方法**来确定项目的**经济可行性**。一个**初始项目投资**为**\$3.0M**，并考虑一个**4 年分析期**。预计在**4 年后**出售该项目，售价为**\$1.6M**。计算预期**净利润**情况下的**回报率**。

End of year	Net profit (\$1,000)
0	0
1	-200
2	-100
3	+500
4	+1,600

Use Eq. (7-2) to transfer the costs from each year to an equivalent present worth. The table below provides an analysis.

End of year	Net profit (\$1,000)	$i = 8\%$ ($P/F, i, n$)	Present value	$i = 6\%$ ($P/F, i, n$)	Present value
0	0				
1	-200	0.9259	-\$185	0.9434	-\$189
2	-100	0.8573	-\$86	0.8900	-\$89
3	+750	0.7938	+\$595	0.8396	+\$630
4	+1,600	0.7350	+\$1,176	0.7921	+\$1,267
		Sum	=\$1,500	Sum	=\$1,619

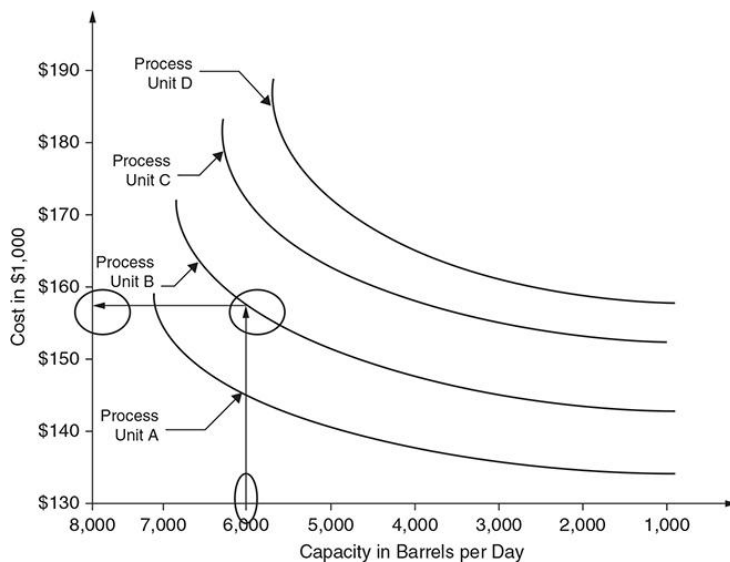
For an interest rate of 8%, the total of the future amounts is equivalent to \$1.5M, which is less than the \$1.6M selling price at the end of the 4-year analysis period. Testing the lower interest rate of 6% yields a total equivalent value of \$1.62M. The rate-of-return for this study is 6%. This rate-of-return should be evaluated against the owner's minimum required rate-of-return.

Additional Examples

E4th_Ch6_Example 6-1

The figure shows cost-capacity curves for process units in a chemical plant. What is the estimated cost for a project that has a process unit B with a capacity of 6,000 barrels per day?

图中显示了化工厂中处理单元的成本-容量曲线。项目中一个 B 处理单元的产能为 6,000 桶/天，请估算其成本。



Locating the 6,000 barrels per day capacity along the abscissa, draw a vertical line upward until it intersects the process unit B cost curve. Then, draw a horizontal line to the left to read the estimated cost as \$158,000.

E4th_Ch6_Example 6-2

The cost of a 375 cubic feet per hour (ft^3/h) process unit is \$5,920,000. From historical cost records, the capacity ratio exponent of a process unit is 0.69. Estimate the cost of a similar process unit with a capacity of 450 ft^3/h .

一个 375 立方英尺/小时 (ft^3/h) 处理单元的成本为 \$5,920,000。根据历史成本记录，处理单元的容量比指数为 0.69。估算一个容量为 450 ft^3/h 的类似处理单元的成本。

$$\begin{aligned}\text{Cost of process unit A} &= (\text{Cost of process unit B}) \times \left(\frac{\text{Capacity of process unit A}}{\text{Capacity of process unit B}} \right)^x \\ &= \$5,920,000 \times \left(\frac{450 \frac{\text{ft}^3}{\text{h}}}{375 \frac{\text{ft}^3}{\text{h}}} \right)^{0.69} \\ &= \$5,920,000 \times 1.13405746 \\ &= \$6,710,000\end{aligned}$$